Find the inverse of each matrix.

11) $\begin{bmatrix} -3 & 1 \\ 9 & -1 \end{bmatrix}$

12) $\begin{bmatrix} -3 & -3 \\ -4 & -3 \end{bmatrix}$

13) $\begin{bmatrix} -4 & 0 \\ -8 & -1 \end{bmatrix}$

14) $\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$

For each matrix state if an inverse exists.

15) $\begin{bmatrix} 2 & -2 & 5 \\ -2 & 3 & -2 \\ 3 & -6 & -4 \end{bmatrix}$

16) $\begin{bmatrix} 2 & -1 & 3 \\ 1 & -1 & -3 \\ -2 & 0 & -4 \end{bmatrix}$

Find the inverse of each matrix.

17) $\begin{bmatrix} -2 & 5 & -2 \\ -2 & 2 & 0 \\ -3 & -2 & 2 \end{bmatrix}$

18) $\begin{bmatrix} 1 & 1 & -2 \\ -3 & -2 & 5 \\ -6 & 4 & 4 \end{bmatrix}$

Critical thinking questions:

19) For what value(s) of x does the matrix M have an inverse?

$$M = \begin{bmatrix} x & 1 \\ 2 & x+1 \end{bmatrix}$$

20) Give an example of a 3×3 matrix that has a determinant of 1.